Food Safety and Analytical Microbiology

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Food Safety and Analytical Microbiology (A00941)

Short Title:	Food Safety & Analytical Micro		
Faculty	Science and Computing		
Department	Science		
Cluster	Food Science		
Author	KGRENNAN		
Credits	10	**Level:**	Postgraduate

Description of Module / Aims

This module will provide the student with an advanced understanding of the theory, practices and implementation issues associated with food industry safety management, while incorporating a comprehensive overview of the theoretical background, methodologies and practical skills in modern methods used in the analysis of foods for microbial contaminants. The learner will be exposed to aspects of working in an environment which operates under strict guidelines for production and quality outputs. The aims of the module are to produce graduates with an in-depth knowledge of food industry safety management, along with modern methods for the microbiological analysis of foods and food production premises. The student will acquire the tools and skills in modern methods and the technology associated with the analysis of foods and production facilities for microbial contaminants. The module will also enable students to critically appraise the scientific literature, and identify future trends in the areas of food safety and analytical microbiology.

Programmes

FOOD-0013	Certificate in Food Safety & Microbiological Analysis (WD_SFOOD_MA)	5 / 0 / M
FOOD-0013	MSc in Analytical Science with Quality Management (WD_SANSC_R)	1 / 0 / E
FOOD-0013	PG Dip in Analytical Science with Quality Management (WD_SANSC_G)	1 / 0 / E

Indicative Content

- Food premises and equipment design
- Clean-room technology; controlled cleaning and structural, procedural and personal hygiene
- Quality control of food ingredients
- H.A.C.C.P.: methodology, documentation, integration and other quality systems (G.M.P., Q-mark, I.S.O. system, T,Q,M., World class manufacturing)
- Food legislation and regulation and organisations involved in food safety (FSAI, WHO, Codex Alimentarius)
- Food safety auditing: theory and practice, the audit process, integrated safety auditing, traceability and complaint management
- Methods for isolation and growth of bacteria and fungi from foods and pharmaceuticals
- Selective enrichment and isolation techniques: chromogenic media & immunomagnetic beads
- Facilitated cultural techniques (Petrifilm, Easygel, Kwikcount and Microtrap)
- Pyrogen detection and endotoxic assays
- Traditional methods of identification of microorganisms: phenotypic methods including microscopy and biochemical tests
- Rapid and/or computer-assisted kit systems for identification (immunological methods, API, Biolog etc.)
- Modern molecular methods for identifying microorganisms: 16S PCR and DNA sequencing, PFGE, AFLP, RFLP, real-time PCR
- Sterilisation methods, sterility testing and monitoring and disinfectant testing; methods for evaluating disinfectants

- Microbiological tests for hygiene testing of production facilities, including ATP and lux-based bioluminescence technology and modern rapid swab techniques

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Evaluate the inter-related processes that contribute to food safety management.
2. Assess the role and function of quality through production and processing of food, including the origins of HACCP and its basis as a Food Safety Management System.
3. Propose suitable techniques for the selective isolation and identification of pathogens and indicator organisms from a wide variety of products using current methodologies and technologies.
4. Evaluate the underlying principles behind molecular biology techniques employed in isolation and identification of microbes.
5. Appraise sterility testing of products and monitoring of sterile production facilities, and recommend appropriate methods for routine hygiene assessment in the food industry.

Learning and Teaching Methods

- Lectures.
- Laboratory practicals.
- Analysis and review of scientific literature.
- Team work.
- Independent learning.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Assignment	50%	3,5
Group Project	10%	2
In-Class Assessment	40%	1,4

Assessment Criteria

<40%: Little or no demonstration of an understanding of the fundamentals of analytical microbiology or the basic concepts of food safety management.

40%-59%: Ability to show limited understanding of the fundamentals of analytical microbiology. Will display some ability with associated demonstrations. Ability to understand the food safety management philosophies but unable to critically think on this subject.

60%-69%: Ability to show reasonable understanding of the complexities of analytical microbiology and predict some trends in analytical results. Will display reasonable competence with associated laboratory demonstrations. Good critical thinking and problem-solving skills in both analytical microbiology and food safety management. Good teamwork abilities.

70%-100%: All the above to an excellent level. In addition, will be able to demonstrate comprehensive knowledge and understanding of the area, have an ability to critically evaluate and relate topics, and formulate food safety management strategies.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		30
Practical		18
Independent Learning		222

Essential Material

- ""Food Safety & Analytical Microbiology" module Moodle page." http://moodle.wit.ie
- "Food Safety Authority of Ireland." http://www.fsai.ie/
- Madigan, Michael T., John M. Martinko, Kelly S. Bender, Daniel H. Buckley and David A. Stahl. _Brock biology of microorganisms_ . 14th ed.. Upper Saddle River, NJ: Prentice Hall/Pearson Education, 2014.

Supplementary Material

- "European Food Safety Authority." http://www.efsa.europa.eu
- Bibek, Ray and Arun Bhunia. _Fundamental food microbiology_ . 4th ed.. Boca Raton, Fla.: CRC; London : Taylor & Francis: Boca Raton : CRC Press, 2008.
- Bisen, Prakash S. . _Microbes: concepts and applications_ . Hoboken, N.J. : Wiley-Blackwell, 2012.
- Doyle, Michael P. and Robert L. Buchanan. _Food microbiology: fundamentals and frontiers_ . 4th ed.. Washington, D.C. : ASM Press, c2013, 2013.
- Mortimore, Sara, Carol Wallace and Christos Cassianos. _HACCP_ . Chichester, GBR : Wiley, 2008.
- Motarjemi , Yasmine and Huub Lelieveld. _Food S__afety Management, A Practical Guide for the Food Industry_. 1st. USA: Academic Press, 2014.
- Pawsey, Rosa, K. _Case Studies in Food Microbiology for Food Safety and Quality_. Cambridge, GBR : Royal Society of Chemistry , 2002.
- Toldra, Fidel and Leo M. L. Nollet. _Advances in Food Diagnostics_. Hoboken, NJ, USA : Wiley-Blackwell , 2009.

Requested Resources

- Room Type: Biology Lab